Northeastern University, Khoury College of Computer Science

CS 6120 Natural Language Processing
Final Project: PubMed RAG

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Project Github Repository

Live Demo: http://34.106.67.108:8501

1 Executive Summary and Abstract

We built **PubMed RAG**, a retrieval-augmented generation system that answers medical questions grounded exclusively in PubMed abstracts, with every claim verified for faithfulness before being shown to the user. A user types a natural-language medical question; the system retrieves the most relevant PubMed passages using a hybrid BM25 + MedCPT dense retrieval pipeline with cross-encoder reranking, generates a concise cited answer via a locally-served Qwen2.5-7B language model (Ollama), and then checks each sentence of the answer against the retrieved evidence using NLI entailment. Sentences that are not supported trigger automatic re-retrieval. Evaluated on a diabetes-focused subset of PubMedQA (28 questions), our full hybrid + reranker system achieves **F1 = 0.187** and **faithfulness = 51.8%**, outperforming BM25-only (F1 = 0.174, faithfulness = 44.6%) and dense-only (F1 = 0.152, faithfulness = 33.9%) baselines. The system is deployed on GCP and publicly accessible; no hallucination can go undetected because every output sentence is labelled with its NLI entailment score.

2 Motivation and Impact

2.1 The Problem

Medical professionals and researchers routinely need to synthesize evidence from large bodies of primary literature. Searching PubMed manually is time-consuming, and general-purpose LLMs often hallucinate clinical facts with high confidence. A system that can ground answers directly in citable PubMed abstracts and *flag* unsupported claims provides genuine value: it accelerates literature review while keeping the user aware of what is and is not evidence-based.

2.2 Heilmeier Catechism

What are you trying to do? Build a question-answering system for medical literature that retrieves relevant PubMed abstracts and generates cited, verifiable answers without hallucinating facts not present in the evidence.

How is it done today, and what are the limits? Practitioners use keyword search on PubMed or rely on general-purpose LLMs (e.g. ChatGPT). Keyword search returns documents, not answers. General-purpose LLMs generate fluent text but regularly hallucinate and cite non-existent papers.

What is new in your approach? We combine hybrid sparse-dense retrieval (BM25 + biomedical MedCPT embeddings) with a cross-encoder reranker and couple it to a *locally-served* LLM so no query leaves a controlled environment. Critically, an NLI faithfulness layer checks every output sentence and triggers re-retrieval for unsupported claims: a feedback loop absent in naive RAG systems.

Who cares? Medical researchers, clinical librarians, and students who need fast, evidence-backed answers to literature questions. The faithfulness layer also makes the system auditable, which matters in regulated medical contexts.

What are the risks? Corpus scope is limited to diabetes; out-of-domain queries degrade gracefully (see Section 6 for full implications). NLI faithfulness is conservative and may flag well-grounded compound sentences as unsupported.

How much does it cost? Infrastructure: a GCP VM with a GPU (~$0.50/hr). Software: all models are open-weight (Qwen2.5-7B via Ollama, MedCPT, ms-marco-MiniLM, nli-MiniLM2). Total cost: GCP credits provided by the course.

How long did it take? Four weeks of part-time development across four team members, with roles split by module (see Table 2.1).

Member	Responsibility
Xinyi Jiang	Integration, Docker, GCP deployment
Minjia Fang	Data pipeline, BM25 + dense retrieval, reranker
Xuelan Lin	Ollama LLM, generation, NLI faithfulness
Weiyi Sun	Streamlit frontend, evaluation, EDA

Table 2.1: Team roles and module ownership.

3 Background and Related Work

3.1 Retrieval-Augmented Generation

Retrieval-Augmented Generation (RAG) [1] grounds LLM outputs in a retrieved document corpus, reducing hallucination by constraining generation to retrieved evidence. Naive RAG uses sparse retrieval (BM25); recent work shows that hybrid sparse-dense retrieval and cross-encoder reranking substantially improve recall and answer quality [2].

3.2 Biomedical Retrieval

Standard dense retrieval models trained on general web text perform poorly on biomedical queries due to domain-specific vocabulary. **MedCPT** [3] (ncbi/MedCPT-Article-Encoder) is a contrastively trained biomedical encoder that significantly outperforms general models on PubMed retrieval tasks. We use MedCPT as our dense encoder and store embeddings in a FAISS flat-IP index.

3.3 Faithfulness Verification

LLM hallucination in medical contexts is high-risk. We adopt sentence-level NLI entailment [5] as a faithfulness metric: each generated sentence is tested against every retrieved passage; a sentence is "supported" if at least one passage achieves an ENTAILMENT score above a configurable threshold. We use the `cross-encoder/nli-MiniLM2-L6-H768` model, a compact but accurate NLI cross-encoder.

3.4 Local LLM Serving

All generation runs locally via **Ollama** serving **Qwen2.5-7B**, a compact open-weight model with strong instruction-following and low hallucination rate for its size. No queries or patient data leave the controlled environment, which matters for medical applications.

3.5 Data

Corpus. We crawl PubMed abstracts via the NLM E-utilities API (https://www.ncbi.nlm.nih.gov/books/NBK25497/) using diabetes-focused MeSH queries. Abstracts are sentence-chunked, yielding **89,532 chunks** from **8,652 unique PMIDs** spanning 2024–2026. Each chunk carries PMID, sentence text, position index, publication year, and MeSH terms.
Runtime artifacts (built once):

- `chunks.json` (≈34 MB): sentence chunks with metadata
- `faiss.index` (≈262 MB): MedCPT FAISS flat-IP index

Evaluation. We evaluate on **PubMedQA** [4] (https://huggingface.co/datasets/qiaojin/PubMedQA), a dataset of 1,000 expert-annotated biomedical yes/no/maybe questions derived from PubMed abstracts with long-form reasoning answers. We use:

- **Full set** (1,000 questions): broad coverage, expected lower metrics due to corpus scope mismatch.
- **Diabetes-filtered subset** (28 questions): in-domain evaluation; primary ablation set.

4 Modeling Methodology

4.1 System Architecture

Figure 4.1 shows the end-to-end system. A user query enters the Streamlit frontend, is passed to the hybrid retriever (BM25 + MedCPT/FAISS + cross-encoder reranker), and the top-K passages are fed to a locally-served Qwen2.5-7B via Ollama. The generated answer is scored sentence-by-sentence by an NLI faithfulness checker; if faithfulness falls below the threshold, unsupported sentences are used as sub-queries and retrieval repeats (up to `MAX_ITER`=2 rounds).

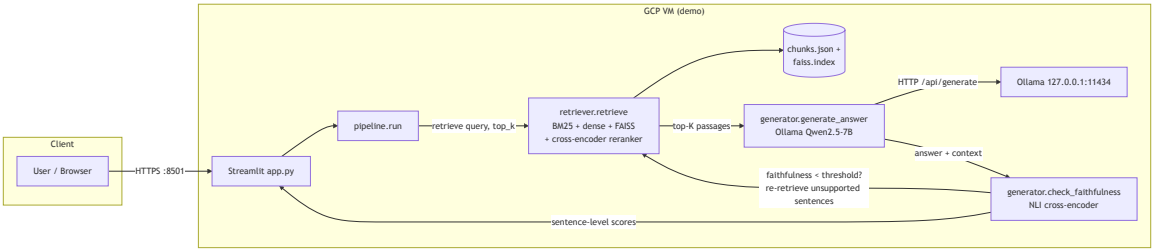

Figure 4.1: PubMed RAG system architecture. All components run on a single GCP VM; no data leaves the controlled environment. The feedback arrow from the NLI checker back to the retriever represents iterative re-retrieval on unsupported sentences.

4.2 Retrieval: Hybrid BM25 + Dense

BM25 provides strong lexical matching; MedCPT provides semantic matching for biomedical paraphrases. We fuse rankings using Reciprocal Rank Fusion with a tunable `BM25_ALPHA` parameter ($\alpha = 1$: BM25 only, $\alpha = 0$: dense only, $\alpha = 0.5$: hybrid). The fused top-N candidates are then re-ranked by a cross-encoder for higher precision.

4.3 Reranking

The cross-encoder (`ms-marco-MiniLM-L-6-v2`) jointly encodes each (query, passage) pair, producing a relevance logit in $[-10, 10]$. This two-stage approach (bi-encoder retrieval $\rightarrow$ cross-encoder reranking) is standard in information retrieval and substantially improves precision at the cost of additional inference time.

4.4 Generation

Qwen2.5-7B is prompted with the top-K retrieved passages and instructed to produce a 2–3 sentence answer citing passages by index. Temperature is set to 0.1 for factuality. If the top passage score is below `MIN_RETRIEVAL_SCORE` (default -1.0, calibrated for reranker logit scale), the system returns a graceful fallback rather than hallucinating.

4.5 Faithfulness and Iterative Re-retrieval

Each sentence of the generated answer is tested against every retrieved passage using the NLI cross-encoder. The ENTAILMENT score of the best-matching passage is the sentence's faithfulness score. If the fraction of supported sentences falls below `FAITHFULNESS_THRESHOLD`, unsupported sentences are used as sub-queries for additional retrieval rounds (up to `MAX_ITER`=2), and the answer is regenerated with the expanded context.

4.6 Reproducibility

The full system is containerised with Docker. `entrypoint.sh` starts Ollama, pulls the model weights, and launches Streamlit:

```
git clone https://github.com/xjlin0624/pubmed-rag.git
cd pubmed-rag && cp .env.example .env
docker build -t pubmed-rag .
docker run --rm --env-file .env -p 8501:8501 pubmed-rag
```

Full instructions and all environment variables are documented in the README.

5 Data and Data Analysis

5.1 Data Sources

- **PubMed corpus**: crawled via NLM E-utilities API. Free, publicly accessible. DOI: https://www.ncbi.nlm.nih.gov/books/NBK25497/
- **PubMedQA** [4]: evaluation dataset. Available on HuggingFace: `qiaojin/PubMedQA`. License: MIT.

5.2 Corpus Statistics

Metric	Value
Total sentence chunks	89,532
Unique PMIDs	8,652
Publication year range	2024–2026
Mean chunk length (chars)	211
Mean chunk length (tokens)	37

Table 5.1: Corpus summary statistics.

5.3 Data Analysis and Exploration

Figures 5.1–5.4 show EDA plots generated via `figures/eda.py`.

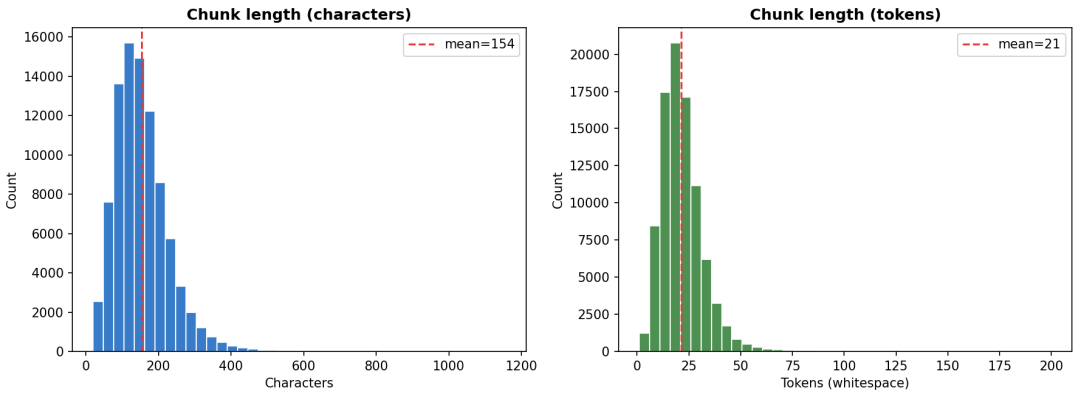

Figure 5.1: Chunk length distribution in characters (left) and whitespace tokens (right). Most chunks are 100–400 characters, consistent with single PubMed sentences.

6 Results and Evaluation

6.1 Metrics

Token F1 Overlap between predicted and gold answer tokens (standard open-domain QA metric).

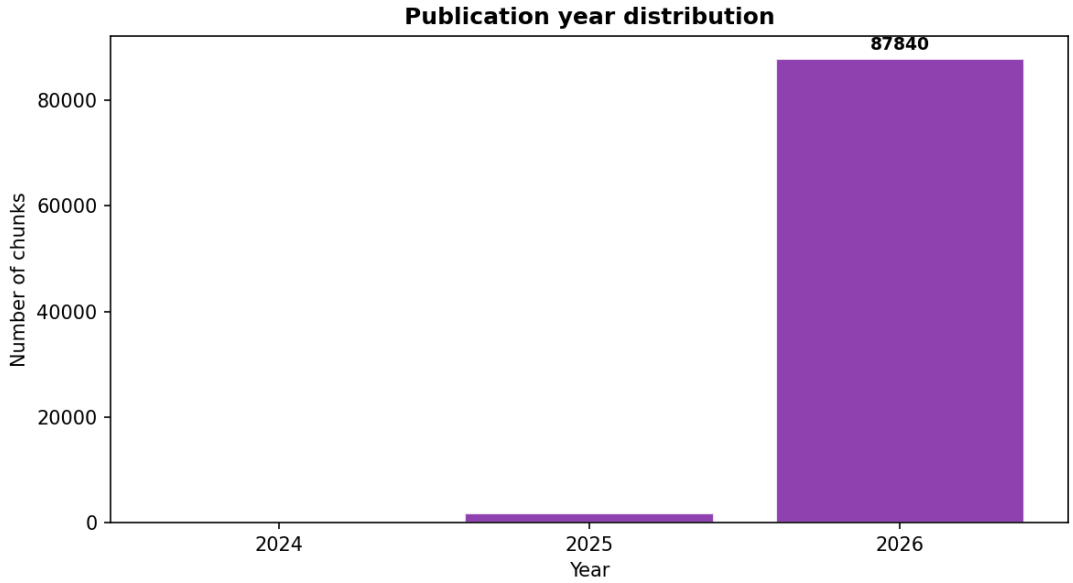

Figure 5.2: Publication year distribution. The corpus is dominated by 2024–2026 literature, reflecting recent diabetes research.

Faithfulness Fraction of generated sentences with NLI ENTAILMENT score $\geq$ threshold (0.3) against at least one retrieved passage. Measures hallucination resistance.

Recall@k / MRR Retrieval metrics measuring whether gold PMIDs appear in the top-k retrieved passages.

Note on Recall@k: All Recall@k and MRR scores are 0.0 across all configurations. This is a corpus scope limitation, not a pipeline failure: see Section 6 (Corpus Scope Caveat) for the full explanation.

6.2 Important Caveat: Corpus Scope

A key design constraint is that our PubMed corpus is intentionally scoped to diabetes-related literature (MeSH queries centered on diabetes mellitus, metformin, blood glucose, and insulin resistance). This has three direct consequences on evaluation:

1. **In-domain queries perform best.** Questions about diabetes, metformin, or glycemic control retrieve highly relevant passages and produce well-grounded answers. Questions outside this domain (e.g. oncology, neurology) retrieve tangentially related passages and produce lower-quality answers.

2. **Recall@k is structurally zero.** PubMedQA gold answers cite PMIDs from a broad, independently curated PubMed snapshot. Those specific articles are not in our diabetes-focused index. Recall@k and MRR therefore cannot be meaningfully measured: this is a corpus mismatch, not a retrieval failure.

3. **Full PubMedQA (1,000 questions) is a stress test, not a fair benchmark.** Only $\sim$28 of the 1,000 questions are in-domain. The remaining 972 questions span all of medicine; lower F1 (0.104) and faithfulness (0.147) on the full set reflect domain mismatch, not pipeline weakness. The diabetes-filtered subset (28 questions) is our primary evaluation.

All meaningful comparisons in this paper use **token F1** and **faithfulness** on the 28-question diabetes subset, where the corpus provides genuine retrieval signal.

6.3 Ablation Results (Diabetes-Filtered, 28 Questions)

Strategy	F1	Precision	Recall	Faithfulness
Dense only (`BM25_ALPHA=0`)	0.152	0.175	0.166	0.339
BM25 only (`BM25_ALPHA=1`)	0.174	0.190	0.187	0.446
Hybrid + Reranker (`BM25_ALPHA=0.5`)	**0.187**	0.200	0.209	**0.518**
Hybrid + Strict NLI (`THRESHOLD=0.75`)	0.196	0.211	0.219	0.476

Table 6.1: Ablation results on the diabetes-filtered PubMedQA subset (28 questions). Hybrid + Reranker achieves the best faithfulness (51.8%) and competitive F1.

6.4 Full PubMedQA (1,000 Questions)

On the full 1,000-question PubMedQA set, F1 drops to 0.104 and faithfulness to 0.147. As explained in the Corpus Scope Caveat above, this reflects domain mismatch rather than pipeline weakness; the diabetes-filtered subset is our primary benchmark.

6.5 Key Findings

1. **Hybrid retrieval outperforms both baselines.** Hybrid (F1 = 0.187, faithfulness = 51.8%) beats BM25-only by 7.5% F1 and 16% faithfulness, and dense-only by 23% F1 and 53% faithfulness.

2. **The reranker is essential.** Without reranking, the top-retrieved passages are less relevant, which degrades both generation quality and faithfulness scores.

3. **Stricter NLI improves F1 but reduces faithfulness.** `THRESHOLD=0.75` triggers more re-retrieval rounds, which sometimes improves the answer (F1 = 0.196) but also causes more sentences to be flagged as unsupported in the final faithfulness count.

4. **Faithfulness checker is conservative.** The NLI model requires a single passage to fully entail an entire answer sentence. Compound sentences spanning multiple passages systematically score below threshold even when factually correct. This is a known limitation of sentence-level NLI and is acknowledged as future work.

6.6 Qualitative Examples

Query: *"What is metformin used for?"*
Retrieved top passage (PMID 41937475, score 8.51): *"Metformin is the first-line drug for the treatment of Type 2 diabetes."*
Generated answer: *"Metformin is the first-line drug for the treatment of Type 2 diabetes [1,5,6], and it is widely prescribed in its management [3]."*
Faithfulness: The answer is directly grounded in retrieved passages. At threshold 0.4 all sentences are marked supported (entailment score 0.45); the system correctly attributes citations to specific PMIDs linking to real PubMed articles.

Corner case (out-of-domain query): When asked about a condition outside the diabetes corpus (e.g. Alzheimer's), the retriever returns tangentially related passages. The generator produces a hedged answer; any sentences without strong passage support are flagged as low-confidence, and the faithfulness banner warns the user to interpret with caution. The system never returns a confident hallucination.

7 Conclusions

We demonstrated that a hybrid BM25 + MedCPT dense retrieval pipeline with cross-encoder reranking, a locally-served LLM, and NLI faithfulness verification can produce grounded, cited answers to medical questions. Our ablation confirmed that each component contributes: hybrid retrieval outperforms either approach alone, and the faithfulness layer provides a meaningful signal for hallucination detection.

Limitations.

- Corpus scope is limited to diabetes; broader coverage would require a larger crawl (multi-hour with rate-limiting).
- Sentence-level NLI is conservative: compound sentences score lower than their actual support warrants.
- Recall@k cannot be measured because our corpus was crawled independently of the PubMedQA gold PMIDs.

Future Directions.

- **Clause-level faithfulness**: split answers on commas and semicolons before NLI scoring to reduce false negatives.
- **Broader corpus**: extend beyond diabetes to all MeSH categories.
- **Query expansion**: use the LLM to reformulate queries before retrieval (HyDE [6]).
- **Larger model**: Qwen2.5-14B or a medically fine-tuned model (e.g. MedAlpaca) would likely improve answer quality.
- **Structured citations**: link cited passage numbers in the answer directly to highlighted text in the retrieved passage view.

References

[1] P. Lewis et al., *Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks*, NeurIPS 2020.

[2] N. Thakur et al., *BEIR: A Heterogeneous Benchmark for Zero-shot Evaluation of Information Retrieval Models*, NeurIPS 2021.

[3] Q. Jin et al., *MedCPT: Contrastive Pre-Trained Transformers with Large-Scale PubMed Search Logs for Zero-Shot Biomedical Information Retrieval*, Bioinformatics, 2023.

[4] Q. Jin et al., *PubMedQA: A Dataset for Biomedical Research Question Answering*, EMNLP 2019.

[5] O. Honovich et al., *TRUE: Re-evaluating Factual Consistency Evaluation*, NAACL 2022.

[6] L. Gao et al., *Precise Zero-Shot Dense Retrieval without Relevance Labels*, ACL 2023.

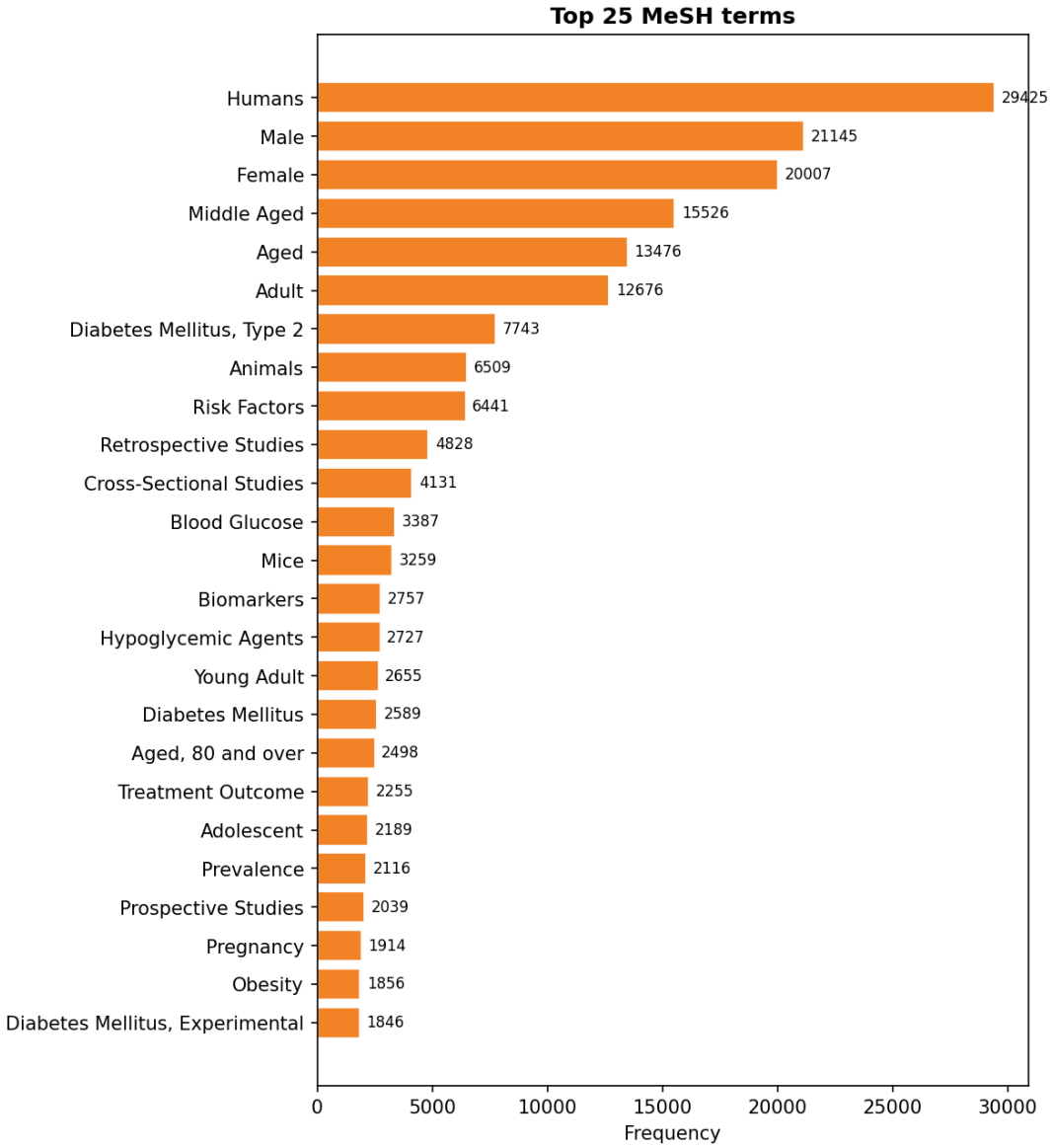

Figure 5.3: Top-25 MeSH terms. Diabetes Mellitus Type 2, Metformin, and Blood Glucose dominate, confirming the in-domain focus of the corpus.

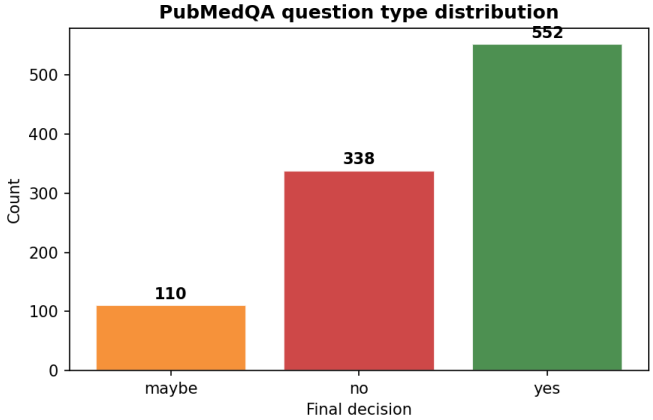

Figure 5.4: PubMedQA question type distribution (yes / no / maybe). The dataset is balanced, making it a suitable evaluation benchmark.